Agenthon Track 2 — independent pre-submission audit

2026-10-16 · tree `hhwolf/phase4-text-integration@ee31323` · toolkit 2.6.0 · vendor `b9c663d` · full report and evidence: `reports/audit/AUDIT.md`

1. Verdict: GO WITH FIXES

Upload the card-notes candidate (numeric_v1.1 + card-notes-only text, λ = 0.5, `models: []`, zero House requests) **after blockers B1–B4 are cleared** and the eight-sweep `t2eval.final_release` verification passes on the exact published digest. **Fallback:** the already public numeric_v1.1 digest `ghcr.io/hhwolf/agenthon-numeric-v1@sha256:f3ec5167b28f0172004a78e336d84a4ecd423d27da42d942693e51cfc085c507`, which passed all four mock-final modes in this audit. **Do not ship LLM labels**: no response cache exists anywhere reachable, so the claimed –0.145 practice gain cannot be reproduced and was never measured on numeric_v1.1 with the integrated overlay.

Today there is **no final artifact**: no final image or digest, no `submission.json`, `release.json` is `pending`, and `text_config.json` is `provisional`.

Option	Practice mean (88)	Caps	Requests	Status today
Card-notes candidate	**2.378141**	9	0	local image only; branch unpushed; no license
numeric_v1.1 public `f3ec5167…` (fallback)	2.415744	9	0	published, anonymous pull OK, mock-final 4/4 modes
LLM labels λ0.5 (teammate report, vs v1)	2.2905	7	≈4–8	unreproducible (no cache)

2. Blockers

#	Blocker	Evidence	Fix
B1	No final image, digest or descriptor; nothing to match across submission.json / CI / mock grader	no `submission.json`; no `final-*` tag or run; `release.json` `pending`	freeze release, tag `final-*`, run `t2eval.final_release` on the digest, generate the descriptor from it
B2	No code license; `license` is a required SPDX field; blocks the fallback too	`LICENSE` missing; schema `required`	human picks a license; add `LICENSE`
B3	Grep gate unmet: 0 of 7 PRs carry any grep review or score; #1–#5 merged without a 5; the candidate branch has no PR	`gh api …/issues/N/comments`, `/reviews`, `/check-runs` all empty	open a PR, iterate to grep = 5 (also for code merged into the fallback)
B4	Candidate exists only locally (9 commits ahead of main, no upstream)	`git status -sb`; `git ls-remote`	push now
B5 (labels only)	No LLM cache: replay ablation impossible	filesystem, branches, artifacts and releases searched	ship labels only if the cache arrives by about Oct 19 and every check passes

3. Key warnings

- **W1** The card-notes gain is partly in-sample: B4 v2 rules were revised after reading practice results (Phase 3), then selected on the same proxies. Expect 0 to –0.03 on the sealed set, not –0.038.
- **W3** Monthly macro 2022 is still 3.2× M0 (0.842 → 2.659); held-out macro is +19% vs M0.
- **W4** Caps never fell (held-out 15 → 18). Practice: 9 F4 caps = 34% of the mean; five sit at 8.3–9.9 unclipped.
- **W5** EM transfer cards are unmeasured. **W6** The eval run-id ignores `TEXT_LAYER_MODE`, so numeric-only and card-notes practice runs overwrite the same directory. **W7** The "cold" pull reused cached base layers.

4. Reproduced comparison (re-run; ⚠ would mark |Δl| > 0.002 vs earlier reports: none)

2,000 draws (M0 also at 500), per-card CRC32 seeds, official scorer. Historical cards have no text, so the final candidate equals numeric_v1.1 there. All 13 reported means reproduce within 5e-7; per-card max |Δl| ≤ 1.8e-15; all cap counts match; 0 failures.

Arm	CV 2009–18 (1,073) mean / caps	Held-out 2019–24 (554)	All hist. (2,060)	Practice (88) mean / median / caps
M0 (500)	1.035657 / 2	1.488258 / 15	1.214629 / 24	2.551735 / 1.3254 / 11
M0 (2000)	1.033389 / 1	1.480838 / 15	1.210785 / 23	2.538247 / 1.3399 / 10
numeric_v1	0.953077 / 1	1.456363 / 17	1.154244 / 23	2.435614 / 1.3770 / 9
numeric_v1.1	0.951081 / 1	1.454208 / 18	1.152537 / 24	2.415744 / 1.3659 / 9
Final (card-notes λ0.5)	0.951081 / 1	1.454208 / 18	1.152537 / 24	**2.378141 / 1.3555 / 9**

Practice by group	n	M0	v1	v1.1	Final	caps
F1	19	0.932	0.848	0.842	0.843	0
F2	23	2.000	2.027	2.034	1.975	0
F3	18	1.376	1.285	1.278	1.277	0
F4	28	4.817	4.588	4.529	4.459	9
horizon 1–21 / 22–63	23 / 28	4.882 / 1.960	4.601 / 2.140	4.526 / 2.141	4.442 / 2.094	7 / 1
horizon 64–126 / 127+	18 / 19	2.138 / 0.932	1.804 / 0.848	1.809 / 0.842	1.803 / 0.843	1 / 0
panel macro_monthly / rates	3 / 39	5.804 / 2.087	3.934 / 2.001	3.876 / 1.999	3.876 / 1.983	1 / 2
panel g10_fx / factors	28 / 13	2.071 / 4.712	2.174 / 4.502	2.150 / 4.438	2.059 / 4.427	1 / 5
shape joint / single	36 / 52	1.502 / 3.256	1.405 / 3.149	1.395 / 3.123	1.394 / 3.059	1 / 8

Text ablation (numeric_v1.1 → card-notes): –0.037603, year-block 95% CI [–0.068127, –0.013183]; F1 +0.0014, F2 –0.0589, F3 –0.0010, F4 –0.0700; caps 9 → 9 (none new or removed); 15/88 cards change. The LLM-label arm could not be run (B5).

5. Acceptance gates

Gate	Result
v1.1 vs v1: CV mean	PASS 0.953077 → 0.951081 (CI [–0.0051, +0.0010])
v1.1 vs v1: practice mean	PASS 2.435614 → 2.415744 (CI [–0.0411, +0.0034])
v1.1 vs v1: CV + practice caps	PASS 10 → 10
No major group > 10% worse	PASS (worst CV +0.38%, practice –0.84%)
Held-out veto > 15%	PASS (worst +0.65%, F2)
Tails variance-normalized	PASS: `np.sqrt(3./rng.chisquare(5.))` (= √(3/5)·t5); mixture `/np.sqrt(1.4)`; FHS normalizes block covariance
Monthly mid-2022 fixed?	**NO.** Macro 2021: M0 5.685 / v1 4.079 / v1.1 4.361; 2022: 0.842 / 2.796 / 2.659
Text rule → card-notes λ = 0.5	PASS: mean improves, caps equal; labels unavailable → correctly *provisional*
Clamps	PASS: drift 0.000 SD on all 88; vol_mult ∈ {1.0, 1.075, 1.125}; never < 1
Post-hoc choices on report-only data	W1 (practice-informed B4 v2, medium); W2 (2019–24 seen in diagnosis, low)

6. Mock Final (empty `DOCKER_CONFIG`; local 4-CPU Colima; 104 units; 88 proxies)

No final digest exists, so the matrix ran on the **public fallback digest** (cold anonymous pull) and on the **candidate image built from** `ee31323` (local, `--allow-local-image`). It must be repeated on the real digest.

Sweep	Pass	Proxy mean	Req. max	Wall mean / max (s)	Σ104 (s)
fallback offline (cold pull 3.7 s)	104/104	2.415744	0	1.30 / 5.66	134.9
fallback restricted canned / chaos / replay	104/104 ×3	2.415744	0	1.35 / ≤1.92	≈140.6
candidate offline / repeat	104/104 ×2	2.378141	0	1.82 / 2.29	189.3 / 184.3
candidate restricted canned / chaos / replay	104/104 ×3	2.378141	0	≈1.87 / ≤2.38	≈194
candidate seeds 1111 / 2222	104/104 ×2	2.379504 / 2.389690	0	1.84 / 2.10	190.9 / 218.8
LLM-labels variant canned (fixture labels, not accuracy)	104/104	2.343591	8	2.89 / 10.29	300.2
LLM-labels variant chaos (12 units, 65 injected-fault requests)	12/12	—	7	122.9 / 124.6	1,474.8

- **All gates PASS for the candidate and the fallback:** 104/104 in every mode; 0 requests; zero non-House egress (internal networks + allowlisting proxy); projected about 190 s per 100 units (0.44% of 43,200 s); output ≤ 424,845 B / 3 files, 0 symlinks, 0 blank rationales.
- **Determinism:** offline vs repeat, and offline vs canned/chaos/replay: 104/104 byte-identical triplets.
- **Fresh seeds:** each mean lies inside the other's bootstrap CI; paired Δ +0.0102, CI [–0.0024, +0.0239]; caps 9/9; every parquet changed (the seed is honored).
- Chaos is vacuous for zero-request images. The House-failure path was exercised with an LLM-labels variant (above): 12/12 units produced valid forecasts; 9 fell back to neutral with the reason recorded in the rationale (`CallFailed: JSONDecodeError … numeric-only`), 3 kept labels after excluding 1 of 3 failed vote runs. No crash.

7. Rules and readiness

linux/amd64 ✔ · non-root `runner` ✔ · no VOLUME ✔ · `/app` = 27 files, 260 KB, allowlisted `.dockerignore` ✔ · no vendor/.eval/realized/.env/keys/LLM cache ✔ · 0/104 unit IDs anywhere in the image ✔ · fallback anonymous manifest + 10 blobs HTTP 200 ✔ · descriptor dry-run via the pinned toolkit (no pack): schema 1.1.0, interface 2.0, forecasting / final / api / public, `models: []` (no learned model) ✔; the House row `nvidia/nemotron-3-super-120b-a12b`, `rl-030326-fp8`, `unpublished`, `api` also validates · license ✘ (B2) · prompts: as-of stated, post-as-of knowledge forbidden, doc_id citations required, no tools, temperature 0, seed 20261014, `max_retries=0`, max_tokens ≤ 4000 ✔.

8. Risk review

Risk	Likelihood	Cost	Mitigation
Text hurts on sealed 2025–26 (memorization)	labels med–high; card-notes low	labels: gain may vanish or reverse; card-notes ≤ +0.02	ship card-notes only
House outage / slowness	n/a for card-notes (0 calls); labels medium	labels: neutral fallback, ≤ 400 s/unit	card-notes removes the dependency
Monthly cards	medium	high per card (2022-type regime)	accept; post-competition fix
EM transfer cards	medium	unknown (never measured)	accept; admissible
10-asset joint	low	low (0.946; 0.40 MiB)	none
Wall-clock overrun	very low (≈190 s total)	catastrophic	labels worst case 100 × 400 s = 93% of the clock: don't ship labels
LLM nondeterminism at rerun	none (byte-identical)	labels: 29% low-agreement units	card-notes

9. Improvements, ranked by gain ÷ risk

Do before upload (everything must end with a full mock-final rerun on the published digest by Oct 22):

#	Action	Effect	Effort	Safe before Oct 23?
1	Push branch, open PR, iterate to grep = 5 (B3, B4)	enables upload	1–4 h	yes
2	Choose and add a license (B2)	enables upload	0.25 h	yes
3	Freeze release, tag `final-*`, publish, run the 8 sweeps (fresh daemon for a true cold pull), write submission.json (B1)	–0.038 practice vs fallback (in-sample; sealed 0 to –0.03)	3–5 h	yes
4	Prepare the fallback descriptor for `f3ec5167…` now	insurance	0.5 h	yes
5	Eval-only: add text env to the run identity (W6)	avoids wrong comparisons	0.5 h	yes (no image change)

Not worth the risk now: LLM labels (cache missing; F4-concentrated –0.43 claim unverified; memorization; House and clock exposure; 10–14 h). F4 family-aware tail floor (could pull five near-cap cards under 8, but it can only be validated on the same practice set). Monthly trend fix (all A4 variants failed; no clean data left).

After the competition: point-in-time macro vintages; regime-switch volatility for quiet-to-moving transitions; EM data; LLM labels tested only on post-training-cutoff episodes; an explicit cap-avoidance objective.

10. Reproduce

```
git checkout ee31323
make test                                  # 241 passed
make admissibility FORECASTER=numeric_v1_1  # 104/104 (audit used --work .context/audit/eval-A)
reports/audit/scripts/run_evals.sh                # generate + 4 backtests + 5 practice arms + admissibility
.venv/bin/python reports/audit/scripts/analyze.py  # comparison-*.csv, gates.json, macro-monthly-by-year.csv
reports/audit/scripts/run_mock.sh                 # 11 mock-final sweeps + labels variant (empty DOCKER_CONFIG)
.venv/bin/python reports/audit/scripts/mock_summary.py
# numeric-only practice arms must use their own --work dir (W6):
TEXT_LAYER_MODE=numeric_only .venv/bin/python -m t2eval.cli eval-practice --forecaster numeric_v1_1 --work
.context/audit/eval-N
# after B1:
.venv/bin/python -m t2eval.final_release --image ghcr.io/hhwolf/agenthon-numeric-v1@sha256:<FINAL> --replay-dir
<dir>
```